PULSE CBD

# Project Timeline

Thirteen-week build schedule, mapped against the project's six software phases and the remaining physical-installation milestones

PREPARED
2026-08-23

STATUS
Software complete & live-verified; physical installation pending

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Structured as thirteen weekly cycles, each combining a brief progress check-in with dedicated project work time. Notably, the full software pipeline — all six build phases, Weeks 4–11 below — is already complete and live-verified against real Transport for NSW data well ahead of the physical-installation milestones. That's a direct consequence of the project's modality choice: the hardware dependency (a physical scale model and projector) is deliberately isolated to a single, late, low-risk calibration step, rather than something the rest of the system has to wait on.

| Wk | Project Stage | Workshop Focus |
|---|---|---|
| 1 | **Project Initiation** | Scoped the problem — making Sydney's live transit network legible and ambient, not just queryable — and surveyed presentation modalities: a fixed 3D-printed/laser-cut CBD model, a Kinect-sensed sand table, and a sensed LEGO grid. |
| 2 | **Exploration & Scoping** | Registered for a Transport for NSW Open Data Hub API key; explored the Complete GTFS static bundle and the GTFS-Realtime feeds across all 7 modes; set the CBD bounding box and surfaced the first feed quirks. |
| 3 | **Proposal Development** | Selected the fixed scale-model modality over the two sensed alternatives; defined the six-phase build order and deliverables; prepared the project proposal. **PROPOSAL DUE** |
| 4 | **Foundation Building** | *Phase 1.* Built the static GTFS ingestion pipeline — spatial filter, SQLite cache — and set up the git repository, config, and automated test suite. |
| 5 | **Data Processing** | *Phases 2–3.* Realtime polling across all 7 modes, protobuf decoding (including TfNSW's custom carriage/occupancy extension), feed-quirk handling, static/realtime entity resolution, map matching, and schedule-adherence scoring. |

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| 6 | **Physical Setup** | *Phase 4.* Stood up the publish layer (WebSocket + HTTP) and the Canvas renderer, verified live on a flat screen as a stand-in for the eventual projector — no physical model exists yet, so this validated the full pipeline end to end regardless. |
| 7 | **Mid-Project Checkpoint** | Live demonstration of vehicles moving on real Sydney Trains, Metro, bus, ferry and light rail data, coloured by live schedule adherence. **MID-PROJECT DEMO + REPORT DUE** |
| 8 | **Hardware Integration** | No depth or stereo sensing to integrate, by design — instead built the calibration tool itself: a draggable-crosshair registration UI and least-squares homography computation, ready for whenever the physical model and projector exist. |
| 9 | **Spatial Mapping** | *Phase 5 continued.* Calibration persisted to disk, automatic switching between the flat-screen projection and a calibrated homography with no renderer code changes, and a verification overlay — tested against synthetic registration points pending real hardware. |
| 10 | **User Experience** | *Phase 6.* Pulsing interchange nodes on the model view, and a separate wall-panel/tablet stats dashboard (adherence, alerts, interchange activity) — deliberately kept off the model surface itself. |
| 11 | **Refinement & Optimisation** | Live testing surfaced and fixed two real bugs: a blocking dialog that froze the renderer, and a WebSocket broadcast loop with no per-client timeout that a single stalled connection could hang for every viewer. |
| 12 | **Final Integration** | Full regression pass (68 automated tests), README and decision log finalised. Physical model fabrication and the real calibration session are the remaining, hardware-dependent steps. |
| 13 | **Project Showcase** | Live demonstration against the real Transport for NSW feeds, projected onto the completed physical model. **FINAL DELIVERABLE + REPORT DUE** |